

ITPROFEL2: OPEN SOURCE PROGRAMMING

Learning Module: Midterm - Week 2

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I. Overview and Objectives

Welcome to Week 2 of Open Source Programming. As senior software development students, you are already familiar with writing code, but this week we pivot to understanding the ecosystem in which modern code lives. We will explore the philosophies that drive open-source software (OSS), the legal frameworks that protect it, and the underlying operating systems that rely on it.

Intended Learning Outcomes (ILOs):

- Define Open Source Software and trace its historical evolution.
- Analyze the practical benefits and technical challenges of adopting Open Source solutions in enterprise environments.
- Examine the critical role of Open Source Licenses in software distribution.
- Define Operating Systems and identify their major types in relation to open-source computing.

II. What is Open Source?

Open source software is software with source code that anyone can inspect, modify, and enhance. "Source code" is the part of software that most computer users don't ever see; it's the code computer programmers can manipulate to change how a piece of software—a "program" or "application"—works. Programmers who have access to a computer program's source code can improve that program by adding features to it or fixing parts that don't always work correctly.

A Brief History

The concept of sharing source code is as old as software development itself. In the 1950s and 1960s, academic researchers and corporate developers frequently shared code to learn from one another. However, the modern open-source movement began in earnest in the 1980s with Richard Stallman's GNU Project and the Free Software Foundation, advocating for users' freedom to run, copy, distribute, study, change, and improve the software. This evolved into the "Open Source" movement in 1998, emphasizing practical business and developmental benefits over strict moral philosophy.

III. Benefits and Challenges of Open Source

Benefits	Challenges
Cost-Effectiveness: Most OSS is free to use, significantly lowering enterprise overhead.	Support and Maintenance: Unlike proprietary software, OSS often lacks a dedicated customer service hotline. You rely on community forums.
Flexibility and Agility: Developers can modify the code to fit specific internal needs without vendor lock-in.	Learning Curve: Setting up and configuring open-source tools can require advanced technical expertise.
Security and Transparency: "Given enough eyeballs, all bugs are shallow." A vast community constantly audits the code.	Orphaned Software: If the community loses interest, the project may be abandoned, leaving security vulnerabilities unpatched.

IV. The Role of Open Source Licenses

Just because software is open source does not mean it belongs to the public domain. Open-source software is tightly bound by licenses that dictate how it can be used, modified, and distributed. Without a license, software is subject to default copyright laws, making it illegal to modify or share.

- **Permissive Licenses (e.g., MIT, Apache 2.0):** Allow users to do almost anything with the code, including using it in closed-source proprietary software, as long as original copyright notices are retained.

- **Copyleft Licenses (e.g., GNU GPL):** Require that any derivative works or modifications also be released under the same open-source license. This ensures the software remains free forever.

V. Operating Systems (OS)

An Operating System is the most critical software that runs on a computer. It manages the computer's memory and processes, as well as all of its software and hardware. It allows you to communicate with the computer without knowing how to speak the computer's language.

Types of Operating Systems

- **Batch Operating Systems:** Users do not interact directly with the computer system. Each user prepares their job offline and submits it to the computer operator.
- **Time-Sharing Operating Systems:** Enables many people, located at various terminals, to use a particular computer system at the same time.
- **Distributed Operating Systems:** Use multiple central processors to serve multiple real-time applications and multiple users.
- **Network Operating Systems:** Runs on a server and provides the capability to manage data, users, groups, security, and networking functions.
- **Real-Time Operating Systems:** Used when time requirements are strict, such as in missile systems or air traffic control.

Note: The open-source movement gained massive traction with the development of the **Linux** kernel, which powers a vast majority of the world's servers, supercomputers, and mobile devices (via Android) today.

Student Activity: The License Breakdown

Instructions for Students:

Form groups of 3-4. Research a popular open-source tool you have used in your previous courses (e.g., VS Code, Git, a specific Python library, or a JavaScript framework). Answer the following questions in a short report:

1. What specific open-source license does this tool use?
2. Is it a Permissive or Copyleft license?
3. If you were to use this tool to build a commercial application that you intend to sell, what restrictions does this license impose on you?

Submit your group report before the next laboratory session.